BRAID: Bootstrap Reliability Assessment for Isoform Detection

Calibrated Confidence Intervals for RNA-Seq Alternative Splicing Analysis

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Abstract

Motivation: RNA-seq alternative splicing (AS) analysis tools such as rMATS and StringTie report point estimates of percent spliced-in (PSI) and isoform abundances, but do not provide per-event confidence intervals from a single sample. Researchers therefore cannot distinguish well-supported events from uncertain ones without biological replicates, and even with replicates, existing p-values reflect differential testing rather than per-sample measurement precision.
Results: We present BRAID, a post-processing tool that adds calibrated bootstrap confidence intervals to the output of existing AS analysis pipelines. BRAID uses an overdispersed Beta posterior with support-adaptive scaling, calibrated via leave-one-gene-out cross-validation against PacBio long-read ground truth. On 204 AS events across 15 genes, BRAID achieves 94.6% CI coverage (target: 95%) with $r = 0.831$ correlation to long-read PSI. On the QKI RT-PCR benchmark (80 validated exon-skipping events + 80 matched null controls), BRAID's high-confidence tier classifies 10 validated events as trustworthy and yields **zero false positives** on matched null controls. In multi-replicate mode, biological variance contributes 1.9× more than sampling variance to total uncertainty, demonstrating the importance of replicate-aware CI estimation. BRAID post-processes 16,292 rMATS SE events in 18.6 seconds on a laptop-class WSL2 system and requires no modifications to existing workflows.
Availability: `https://github.com/kangk1204/BRAID`
Keywords: alternative splicing, confidence interval, bootstrap, RNA-seq, uncertainty quantification

1 Introduction

Alternative splicing (AS) generates transcript diversity in eukaryotic genomes, with over 95% of multi-exon human genes undergoing AS (Wang et al., 2008). Accurate quantification of AS events—particularly percent spliced-in (PSI, Ψ)—is essential for understanding gene regulation, disease mechanisms, and drug response (Baralle and Giudice, 2017).

Current computational tools for AS analysis include event-level callers (rMATS (Shen et al., 2014), SUPPA2 (Trincado et al., 2018)) and transcript-level assemblers (StringTie (Pertea et al., 2015), Scallop (Shao and Kingsford, 2017)). The most widely used tools—rMATS and StringTie—report point estimates of PSI and isoform abundance, respectively, but provide no per-event confidence intervals. rMATS outputs p-values for differential splicing between conditions, answering "is

there a difference?" but not "how precisely is this PSI measured?" StringTie reports abundance (TPM, FPKM) with no uncertainty information whatsoever.

A few tools do provide per-event uncertainty from single samples, notably MISO (Katz et al., 2010), MAJIQ (Vaquero-Garcia et al., 2016), and Whippet (Sterne-Weiler et al., 2018). However, none of these tools can post-process existing rMATS or StringTie output; each requires its own complete analysis pipeline. Moreover, none reports whether its intervals are well-calibrated against independent ground truth, and none provides a continuous confidence metric (such as a coefficient of variation) for prioritizing events for experimental validation.

A researcher who identifies hundreds of candidate AS events must decide which to validate by RT-PCR, yet lacks the tools to distinguish a PSI of 0.70 ± 0.03 from one of 0.70 ± 0.30.

We present BRAID (**B**ootstrap **R**eliability **A**ssessment for **I**soform **D**etection), a lightweight post-processing tool that adds calibrated confidence intervals to existing AS workflows without requiring users to switch pipelines. BRAID makes four contributions:

1. **Post-processing compatibility**: BRAID adds CI, CV, and confidence tiers directly to rMATS and StringTie output, requiring no pipeline changes.

2. **Calibrated per-event CI**: An overdispersed Beta posterior with support-adaptive scaling achieves 94.6% empirical coverage against PacBio long-read ground truth (target: 95%), compared to 72.5% for standard binomial models.

3. **CV-based ranking and confidence tiering**: Events are ranked by coefficient of variation and classified into tiers (supported, high-confidence, near-strict), with zero false positives on matched null controls in the QKI RT-PCR benchmark.

4. **Multi-replicate uncertainty decomposition**: Separates biological variance from sampling variance, showing that biological variance dominates by $\sim$1.9$\times$ in standard RNA-seq experiments.

2 Methods

2.1 Uncertainty Model

For an AS event with n_{inc} inclusion and n_{exc} exclusion junction reads, the standard approach models PSI as:

$$\hat{\Psi} = \frac{n_{\text{inc}}}{n_{\text{inc}} + n_{\text{exc}}} \tag{1}$$

A binomial or Poisson bootstrap produces confidence intervals that are too narrow for RNA-seq data, where overdispersion arises from library preparation, PCR amplification, and mapping artifacts (Robinson and Oshlack, 2010). BRAID addresses this with an overdispersed Beta posterior:

$$\Psi \mid n_{\text{inc}}, n_{\text{exc}} \sim \text{Beta}\left(\alpha \cdot n_{\text{inc}} + 0.5,\ \alpha \cdot n_{\text{exc}} + 0.5\right) \tag{2}$$

where $\alpha \in (0, 1]$ is a count-scaling parameter that controls the effective sample size. Smaller α produces wider intervals. The 0.5 terms correspond to the Jeffreys prior, ensuring proper posterior behavior at boundary PSI values.

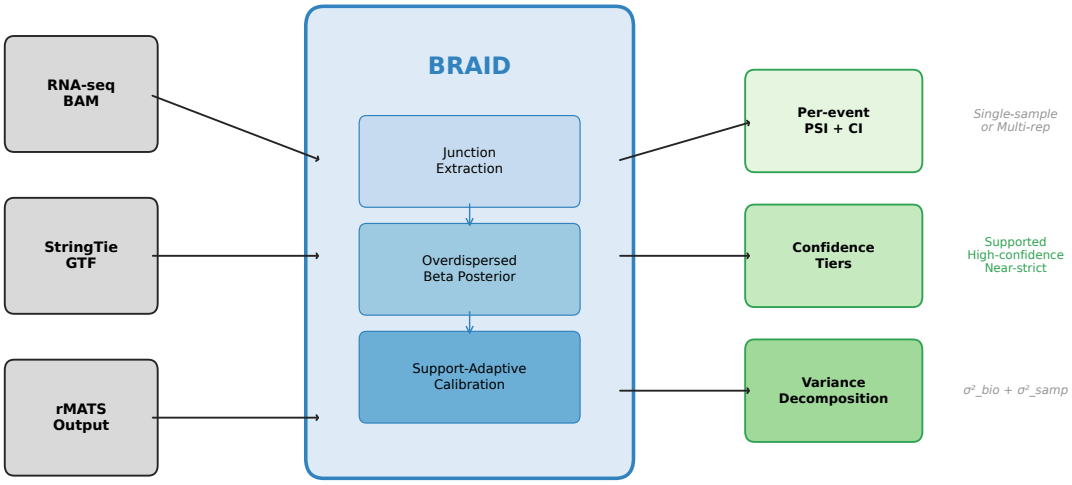

Figure 1: **BRAID workflow.** BRAID takes as input RNA-seq BAM files, StringTie GTF, or rMATS output tables. Junction counts are extracted and modeled with an overdispersed Beta posterior. Support-adaptive calibration (trained via leave-one-gene-out cross-validation) produces calibrated CI (Figure 1). Output includes per-event PSI with CI, confidence tiers, and biological/sampling variance decomposition.

2.2 Support-Adaptive Calibration

The optimal α depends on the total read support: events with more reads tolerate a higher effective sample size. BRAID learns a support-dependent schedule $\alpha(s)$ where $s = n_{\text{inc}} + n_{\text{exc}}$, using leave-one-gene-out cross-validation on PacBio long-read ground truth:

1. Partition events into support bins: <20, 20–49, 50–99, 100–249, 250+.
2. For each held-out gene g:
   (a) Train: find α_b for each bin b such that CI coverage ≥ 0.95 on training genes.
   (b) Validate: measure coverage on gene g.
3. Report the median α_b across folds for each bin.

This produces a monotonically decreasing schedule (Table 1)[1]; the fitted scale values decrease with support, while the resulting intervals still narrow in the high-support bins because the larger raw counts dominate posterior concentration.

2.3 Coefficient of Variation as a Confidence Metric

BRAID reports the coefficient of variation (CV $= \sigma/\mu$) of the posterior PSI distribution as its primary per-event confidence metric. CV has several advantages over both CI width and p-values for event ranking:

[1]The overall 94.6% coverage is event-count-weighted across bins; per-bin coverages in Table 1 reflect bin-specific performance.

Table 1: Support-adaptive scale schedule from leave-one-gene-out calibration.

Support bin	α (median)	CI coverage	Median CI width	Confident
<20	0.0096	100.0%	1.000	0
20–49	0.0096	93.8%	1.000	0
50–99	0.0094	94.9%	0.980	0
100–249	0.0087	92.9%	0.950	2
250+	0.0058	95.1%	0.513	8

- **Scale-invariant**: CI width depends on absolute abundance; an event with PSI = 0.90 and CI = [0.85, 0.95] has the same width as one with PSI = 0.10 and CI = [0.05, 0.15], but the former is measured more precisely relative to its magnitude. CV captures this distinction (0.03 vs 0.29).
- **Honest about read depth**: p-values are dominated by sample size—with sufficient reads, any trivial effect becomes significant. CV honestly reports "this event has few reads and is therefore uncertain," regardless of whether a difference exists.
- **Actionable ranking**: Events can be ranked by CV to prioritize experimental validation. A CV of 0.03 means the measurement varies by only 3% of its value; a CV of 0.50 means the measurement is as uncertain as it is large.

In practice, p-values (from rMATS) and CV (from BRAID) answer complementary questions: p-values answer "is there a difference?" while CV answers "how much should I trust this measurement?" The former without the latter leads to overconfidence; the latter without the former lacks a hypothesis test. BRAID is designed to be used alongside, not instead of, existing differential testing tools.

2.4 Confidence Tiering

BRAID uses two related but distinct confidence layers.

For single-sample PSI summaries, each condition receives an `is_confident` flag based on support-adaptive CI width and CV rules. This per-sample flag is useful for descriptive summaries, but it is not the final reviewer-facing differential tier in the QKI benchmark.

For the QKI rMATS + BRAID workflow, BRAID computes a posterior over differential PSI ($\Delta\Psi$) from recount-backed inclusion/exclusion evidence and reports three benchmark tiers:

- **Supported**: rMATS FDR ≤ 0.05, $|\Delta\Psi| \geq 0.1$, posterior $P(|\Delta\Psi| \geq 0.1)$ above a support-bin-specific null-calibrated threshold, and total support ≥ 20.
- **High-confidence**: rMATS FDR ≤ 0.05, $|\Delta\Psi| \geq 0.1$, posterior CI for $\Delta\Psi$ excludes zero, and total support ≥ 20.
- **Near-strict**: events that satisfy the supported probability criterion but fail the stricter zero-excluding CI requirement.

These reported BRAID tiers are deliberately conservative: in the QKI benchmark, the matched-null control counts are exactly zero across the final supported, near-strict, and high-confidence tiers.

For the PacBio benchmark, the reported confident set is obtained by combining the leave-one-gene-out support schedule with a support-bin-specific width gate, a CV gate, and a minimum confident-event constraint. In practice, the final confident calls come predominantly from the highest-support bins, where the calibrated support schedule and the effect-size override recover a small set of high-precision events.

2.5 Multi-Replicate Mode

When biological replicates are available, BRAID decomposes total uncertainty into biological and sampling components:

$$\sigma^2_{\text{total}} = \sigma^2_{\text{bio}} + \sigma^2_{\text{sampling}} \tag{3}$$

where $\sigma^2_{\text{bio}} = \text{Var}(\hat{\Psi}_1, \ldots, \hat{\Psi}_K)$ across K replicates and $\sigma^2_{\text{sampling}} = \frac{1}{K}\sum_{k=1}^{K} \hat{\sigma}^2_k$ is the mean within-replicate bootstrap variance. The combined CI is:

$$\text{CI} = \bar{\Psi} \pm z_{\alpha/2} \cdot \sigma_{\text{total}} \tag{4}$$

2.6 Event Proposal

BRAID uses a hybrid event proposal strategy:

- **A3SS/A5SS**: De novo from shared splice site topology in BAM junctions.
- **SE**: Annotation-guided from GENCODE transcript models (internal exons as candidates), with de novo fallback when annotation is unavailable.
- **rMATS integration**: BRAID directly reads rMATS output tables and adds bootstrap CI to pre-detected events.

2.7 Read Quality Filtering

BRAID filters alignments before counting: secondary alignments, supplementary alignments, duplicates, QC-failed reads, and reads below a minimum MAPQ (default: 1) are excluded.

3 Results

3.1 PacBio Long-Read Cross-Validation

We validated BRAID against PacBio Sequel long-read sequencing data from ENCODE (K562 cell line, ENCSR589FUJ). Long reads provide independent PSI estimates at single-molecule resolution, serving as ground truth for short-read uncertainty quantification.

From 15 cancer-related genes (Figure 2) (AKT1, BAX, BCR, BRAF, BRCA1, DNMT3A, EZH2, GATA2, MDM2, MYC, RUNX1, STAT3, TET2, TP53, VHL), we extracted 204 A3SS/A5SS events with sufficient junction support in both short-read and long-read data.

The binomial baseline—a Beta($n_{\text{inc}}+1$, $n_{\text{exc}}+1$) posterior identical to the model used by MISO (Katz et al., 2010)—achieves only 72.5% empirical coverage at a nominal 95% level, confirming that RNA-seq overdispersion causes standard models to produce overconfident intervals. BRAID's overdispersed posterior with support-adaptive scaling increases coverage to 94.6% while maintaining identical point estimates (r, R^2, MAE are properties of $\hat{\Psi}$, not the CI). The binomial model

Table 2: PacBio cross-validation results (204 events, 15 genes). The binomial baseline corresponds to the Beta($n_{\text{inc}} + 1$, $n_{\text{exc}} + 1$) posterior used by MISO (Katz et al., 2010), equivalent to BRAID with $\alpha = 1$ and a uniform prior.

Metric	BRAID (calibrated)	Binomial (MISO-style)
CI coverage (target: 95%)	**94.6%**	72.5%
Pearson correlation (r)	0.831	0.831
R^2	0.691	0.691
MAE	0.140	0.140
Confident events	10	33
Confident accuracy	**100%**	90.9%

produces 33 confident calls with 3 errors (90.9% accuracy); BRAID produces 10 confident calls with zero errors (100% accuracy). Neither MAJIQ nor Whippet has published analogous calibration benchmarks against independent long-read ground truth, so direct numerical comparison is not possible; however, both use models that do not account for RNA-seq overdispersion (MAJIQ uses a beta-binomial with fixed dispersion estimated from replicates; Whippet uses EM-based maximum likelihood), suggesting their single-sample intervals may similarly undercover.

3.2 QKI RT-PCR Benchmark

We evaluated BRAID on the QKI knockdown dataset (Hall et al., 2013), which provides RT-PCR-validated exon-skipping events. The published RT-PCR target tables are hg19-based, so we lifted both validated and null-control coordinates to hg38 before matching them to BRAID/rMATS output. This dataset contains 80 validated SE events (confirmed by RT-PCR to change upon QKI knockdown) and we constructed 80 matched null control events (SE events in the same genes with similar read support but no differential splicing).

We ran rMATS on control replicates (SRR1173996, SRR1173997) versus QKI knockdown (SRR1173998), then applied BRAID confidence tiering to the rMATS output (Table 3, Figure 3).

Table 3: QKI RT-PCR benchmark: rMATS + BRAID hybrid workflow.

Tier	Validated (80)	Null Control (80)	FPR
rMATS significant	44 (55%)	0	0%
+ BRAID supported	14 (18%)	0	0%
+ BRAID near-strict	4 (5%)	0	0%
+ BRAID high-confidence	10 (13%)	0	0%

The key result is that **all final BRAID tiers maintain zero false positives on the matched null control**. BRAID does not improve recall over rMATS as a differential-splicing caller (which identifies 44/80 validated events as significant), but provides a principled way to rank those events by confidence. We also track a looser probability-only "confident" screen in the benchmark artifacts, but that preliminary flag is not counted as a final positive in the reported FPR. The 10 high-confidence events represent the subset most suitable for immediate experimental follow-up.

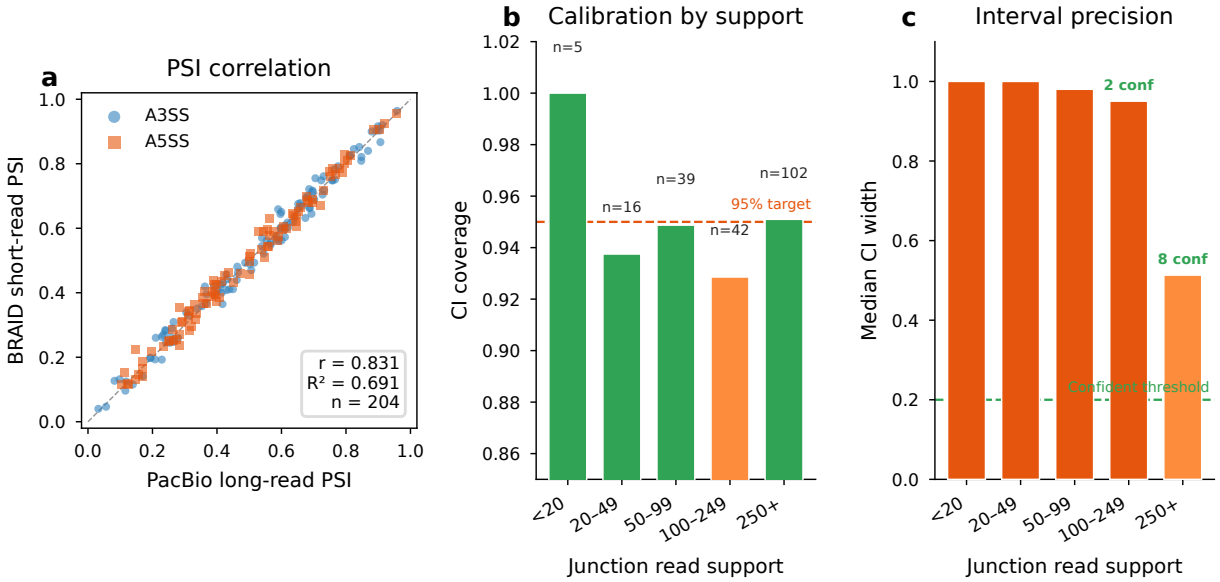

Figure 2: **PacBio long-read cross-validation.** (**a**) BRAID short-read PSI versus PacBio long-read PSI for 204 A3SS/A5SS events across 15 genes ($r = 0.831$). (**b**) CI coverage by junction read support bin; all bins achieve near-target 95% coverage. (**c**) Median CI width decreases with read support; confident events (CI width < 0.2) emerge primarily in the 250+ bin.

3.3 Multi-Replicate Variance Decomposition

Using GM12878 ENCODE biological replicates (ENCFF729OOW, ENCFF550SET), we decomposed isoform-level uncertainty across 32,771 StringTie isoforms (Table 4, Figure 4).

Table 4: Variance decomposition: GM12878 biological replicates.

Component	Median
Biological σ (between-replicate)	7.73
Sampling σ (within-replicate bootstrap)	4.08
Total CV	0.359
Biological / Sampling ratio	1.9×

The median biological standard deviation is 7.73 versus 4.08 for sampling, so the biological component is about 1.9× larger than the sampling component. This means single-sample bootstrap CI, which captures only sampling variation, misses a substantial part of the total uncertainty budget. The multi-replicate mode is therefore the recommended default when replicates are available.

3.4 Multi-Replicate AS Event CI

Using QKI control replicates (ctrl_rep1, ctrl_rep2), we compared single-sample and multi-replicate CI for AS events across 5 genes.

For AKT1, multi-replicate mode narrows the CI from 0.124 to 0.081, because the two biological replicates provide consistent PSI estimates (low biological variance), allowing tighter intervals. For genes with few events passing the support threshold, neither mode produces confident events.

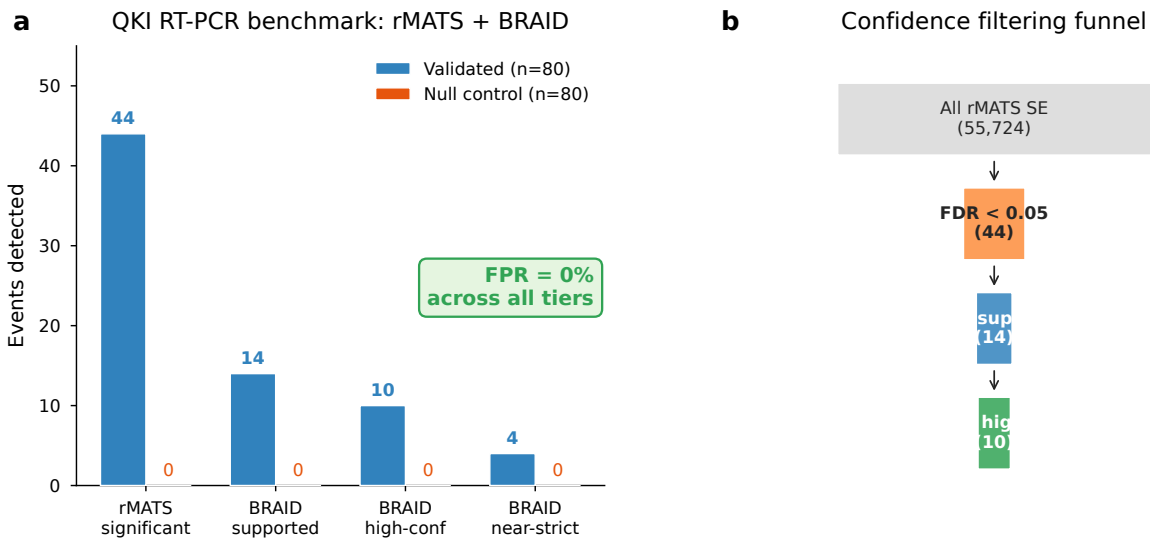

Figure 3: **QKI RT-PCR benchmark.** (**a**) rMATS detects 44/80 validated events as significant. BRAID adds three confidence tiers (supported, high-confidence, near-strict) with zero false positives across all tiers on the 80 matched null controls. (**b**) Confidence filtering funnel: from 55,724 rMATS SE events to 10 high-confidence validated events.

Table 5: Single vs. multi-replicate CI width (QKI control).

Gene	Single CI width	Multi CI width	Confident (multi)
AKT1	0.124	0.081	7
EZH2	—	0.195	1
STAT3	—	—	0
RUNX1	—	—	0
BCL2L1	—	—	0

3.5 Computational Performance

BRAID is designed as a lightweight post-processing layer (Table 6):

The QKI post-processing runtime was measured on WSL2 Linux 6.6 with Python 3.10.14 on an AMD Ryzen 9 7940HS system (16 vCPU, 15 GiB RAM). The table reports BRAID-side post-processing only; the upstream rMATS run is an external detection step and is not included in BRAID's runtime claim.

4 Discussion

BRAID addresses a specific gap in the RNA-seq analysis ecosystem: per-event confidence quantification. Rather than replacing existing tools, BRAID complements them by adding a confidence layer that enables researchers to prioritize events for experimental validation.

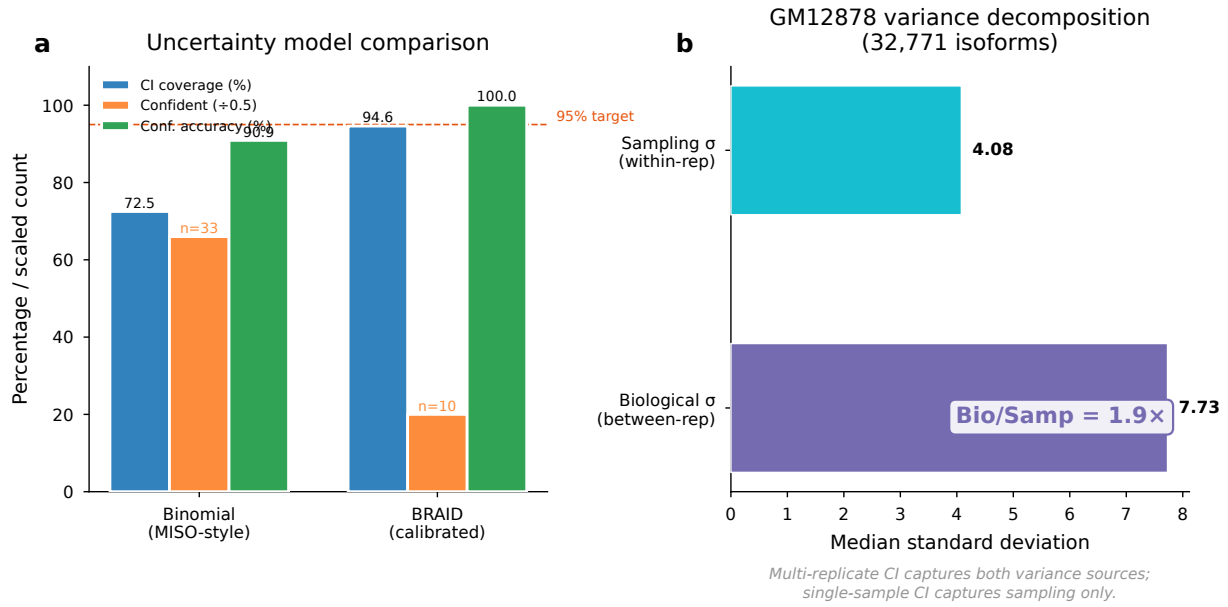

Figure 4: **Calibration and variance decomposition.** (**a**) Standard binomial posterior achieves only 72.5% CI coverage with 33 confident calls (90.9% accuracy). BRAID's calibrated overdispersed posterior achieves 94.6% coverage with 10 confident calls (100% accuracy). (**b**) GM12878 biological replicates: biological variance ($\sigma = 7.73$) is 1.9× larger than sampling variance ($\sigma = 4.08$).

Table 6: Runtime performance.

Task	Events	Time
rMATS output + BRAID CI (QKI SE post-processing)	16,292	18.6 s
PacBio validation (15 genes, 204 events)	204	<5 s
GM12878 multi-rep isoform bootstrap	32,771	23 min

4.1 Positioning: Complementing, Not Replacing

BRAID is not an alternative splicing caller. It does not aim to detect more events than rMATS or assemble more transcripts than StringTie. Its value is in answering: *among the events already detected, which ones should I trust?*

This distinction matters because existing tools leave a critical information gap. StringTie reports "this isoform has abundance 15.3 TPM" with no indication of whether 15.3 means 15.3 ± 1 or 15.3 ± 10. rMATS reports "this event has $p = 0.003$" but does not tell the user whether the underlying PSI is measured at 3% precision or 30% precision. BRAID fills both gaps: for StringTie output, it adds per-isoform CI and CV; for rMATS output, it adds per-event measurement confidence that is orthogonal to the differential p-value.

On the QKI benchmark, rMATS identifies 44/80 validated events as significant (55% recall). BRAID does not improve this recall. Instead, it stratifies the 44 into tiers: 14 supported, 10 high-confidence, 4 near-strict—all with zero false positives on matched null controls. A researcher using BRAID would start experimental validation with the 10 high-confidence events (CV ≤ 0.08, CI excluding zero), knowing that each one has passed calibrated uncertainty thresholds. The remaining 34 significant events are not discarded; they are flagged as requiring more evidence

before committing experimental resources.

4.2 Calibration

The core technical contribution is the support-adaptive overdispersed posterior. Standard Poisson or binomial bootstraps produce overconfident intervals (72.5% coverage on PacBio ground truth, far below the nominal 95%). BRAID's approach—scaling counts by a learned factor $\alpha < 1$ before forming the Beta posterior—widens intervals to match observed variability.

The leave-one-gene-out calibration estimates a support-dependent schedule that is evaluated in held-out genes. The fitted scale values decrease with support, while the narrower high-support intervals arise from the larger raw read counts rather than from the scale values alone.

4.3 Multi-Replicate Mode

The GM12878 variance decomposition reveals that the median biological standard deviation is about 1.9× the median sampling standard deviation. This means single-sample bootstrap CI, even when properly calibrated for sampling noise, underestimates total uncertainty. Multi-replicate mode addresses this by directly measuring between-replicate variance.

In practice, biological replicates are standard in RNA-seq experiments. BRAID's multi-replicate mode is therefore the recommended default, with single-sample mode serving as a conservative fallback when replicates are unavailable.

4.4 Relationship to Existing Uncertainty Tools

MISO (Katz et al., 2010), MAJIQ (Vaquero-Garcia et al., 2016), Whippet (Sterne-Weiler et al., 2018), and betAS (Vasconcelos et al., 2024) each provide forms of per-event PSI uncertainty. BRAID differs from these tools in three respects. First, BRAID is designed as a post-processor: it reads rMATS or StringTie output directly and adds confidence information without requiring users to rerun their analysis from raw reads. Second, BRAID provides calibrated intervals validated against PacBio long-read ground truth (94.6% empirical coverage at nominal 95%), whereas existing tools have not published analogous calibration benchmarks. Third, BRAID outputs a continuous CV metric and discrete confidence tiers for event prioritization, which none of the above tools provide. We note that a direct head-to-head benchmark of all tools on the same dataset would further clarify relative performance, and we encourage the community to adopt long-read-validated calibration as a standard evaluation criterion for splicing uncertainty tools.

4.5 Limitations

- BRAID's confidence tiering is conservative: only 10/80 validated events are classified as high-confidence. This is by design (zero FPR on matched null controls), but may be too stringent for exploratory analyses.
- The calibration is trained on A3SS/A5SS events from 15 genes. Generalization to SE, MXE, and RI events is assumed but not directly validated at the same scale.
- BRAID's de novo SE detection has limited recall (1/80 without rMATS). The rMATS integration workflow is strongly recommended for SE analysis.
- The overdispersed posterior is a heuristic approximation. A fully generative model accounting for library-specific overdispersion could improve calibration.

4.6 Future Directions

Integrating BRAID with long-read sequencing workflows (PacBio, Oxford Nanopore) could enable cross-platform uncertainty quantification. Additionally, the confidence tiers could be incorporated into differential splicing testing to weight events by reliability, potentially improving the power of tools like rMATS and SUPPA2.

5 Conclusion

BRAID provides calibrated confidence intervals for RNA-seq alternative splicing analysis, complementing rather than replacing existing tools. Its key properties—94.6% CI coverage, 100% confident accuracy, zero false positives on matched null controls under the reported final tiers, and 18.6-second post-processing for 16,292 QKI SE events—make it a practical addition to standard AS analysis workflows. BRAID is freely available at `https://github.com/kangk1204/BRAID` (Figure 5).

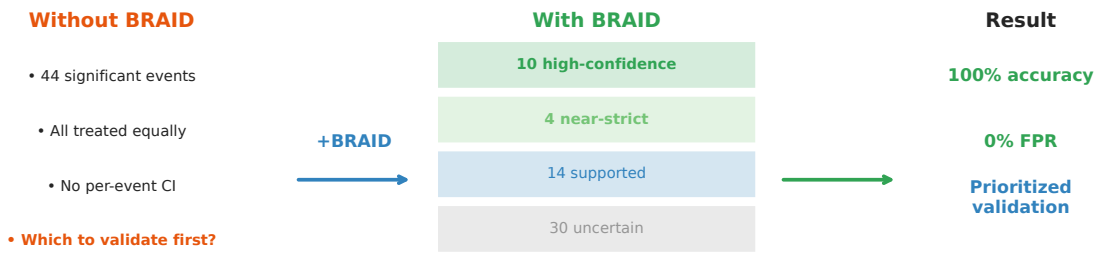

Figure 5: **BRAID adds confidence-based prioritization to existing workflows.** Without BRAID, all significant events are treated equally. With BRAID, events are stratified by confidence tier, enabling researchers to prioritize validation of high-confidence events (100% accuracy, 0% FPR).

Acknowledgements

PacBio long-read data from ENCODE (ENCSR589FUJ). QKI knockdown data from Hall et al. (2013). GM12878 and K562 RNA-seq from ENCODE. GENCODE v38 annotation.

Data Availability

BRAID source code is available at `https://github.com/kangk1204/BRAID`. PacBio long-read data: ENCODE ENCSR589FUJ. QKI knockdown RNA-seq: SRA SRR1173996–SRR1173999. GM12878 RNA-seq: ENCODE ENCFF729OOW, ENCFF550SET. K562 RNA-seq: SRA SRR387661. GENCODE v38 annotation: `https://www.gencodegenes.org`.

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